SRCNN on KV260: experiment evolution：Phase 4: ROM, clock and T4 workstream

Condensed from 231 registered experiments (results.csv). Only turning points are drawn; every ID links to experiments/<ID>_*/. Numbers are ms/frame; the evidence level is stated on each node.

Metric: ms/frame = cycles x clock period. HLS = HLS estimated cycles x target clock; RTL x routed = C/RTL co-simulation cycles x an independently closed post-route clock. No value is a board measurement.

Solid: baseline or parameter evolution, retry after fix, integration of an existing module; dashed: numerical evidence or inspiration, control, verification of the same design, replay. Edges are design lineage, not Git history.

passed, comparable metric (★ selected) · passed with warning · failed or below gate · invalid configuration · timeout or interrupted · planned, not run · analysis · partial experiment, no whole-design metric

P01 Packing PE: two 8-bit products per DSP

E065 Conv2-only 8-bit weights

W21 Sub-kernel cache layout

C13 T4 with pre-packed Conv2

Fold T=4; Conv2 uses E065's 8-bit weights packed two per DSP once per frame

HLS 1.05 ms, 1,032 DSP

! HLS LUT 127,966 is over the old 70% HLS gate but inside the routed screen

F47 Window FIFO 2 -> 16

C23 Gated parameter reload

Load parameters once, reuse across frames

HLS 1.05 ms

! RTL (C27) showed a 4-byte parameter read on a call that should skip loading

C17 C13 + 32-entry window FIFO

Only the FIFO depth, reusing F47's evidence at T4

HLS 1.05 ms

! HLS estimate; RTL and routing needed

C18 Implement C13 at 5 ns

RTL 300,632 cycles (C14); routed LUT 58,529, WNS +0.047; user asked for this route first

RTL x routed 1.50 ms

C28 Reload guard fix

Stop HLS from hoisting a scalarized bias read

RTL (C31): four calls exact, zero skipped-call AXI

★ C22 Implement C17 at 5 ns

RTL 273,023 cycles (C20); routed at 5.0 ns, WNS +0.013

RTL x routed 1.365 ms

C32 C17 + guarded reload

Transplant only the reload interface into C17

RTL 273,025 cycles (C35)

✗ Implementation C36 fails 5 ns setup: WNS −0.102

WSI System integration and board test

Wrap the verified C22 IP with DDR and driver; measure end-to-end on the KV260

not run

W36 Implement C17 at 4.5 ns

Same RTL, tighter clock

✗ Setup fails: WNS −0.216 ns, worst paths are Winograd arithmetic with 8 to 11 logic levels

C38 Balanced registered adder tree

Four registered fabric-add levels in the Winograd fold

✗ HLS FF 248,672 exceeds the 230 k screen; stopped before RTL

C37 Registered parameter packet

Add one 224-bit register stage between parameter FIFO and banks

RTL 273,025 cycles (C39)

✗ Implementation C40 at 4.5 ns fails: WNS −0.278